DEV PATEL

Senior Full-Stack Engineer · Cloud Platform · Event-Driven Systems · AI Agent Platforms
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SUMMARY

Senior full-stack engineer with 5 years of experience building cloud platforms, real-time applications, and production AI systems. Set the architecture, infrastructure-as-code, authentication, and frontend standards supporting 80+ backend services and 15+ React applications. Architected event-driven AWS systems processing 1M+ events a day and built an Amazon Bedrock agent platform that turned week-long investigations into self-serve answers.

EXPERIENCE

Glidewell Dental — Irvine, CA September 2021 – Present

Software Engineer III (2025 – Present) · Software Engineer II (2022 – 2025) · Software Engineer I (2021 – 2022)

Joined as the first software engineer on a hardware engineering team bringing connected devices online across five sites and built the cloud telemetry backend and the application platform above it. Own architecture and delivery end to end — design the systems, deploy the infrastructure, then guide two engineers through implementation and review changes before production.

APPLICATIONS & FRONTEND PLATFORM

- Established the shared React 19 / MUI / Vite design system used across the application portfolio — token-driven light/dark theming, a layout shell, and a reusable component kit (command palette, data grid, dialogs, status chips, empty states) — so a new app starts from a working, consistent shell instead of a blank page.
- Replaced hours of daily manual status checks with 15+ real-time operational dashboards — live KPI, throughput, utilization, and exception views streamed over WebSocket to teams across five sites and nine production lines.
- Digitized a paper-based production process by building a configurable workflow engine — ordered steps with state transitions, conditional branching, scripted actions, and inventory side-effects — so every step emits structured events, resurfaced as LLM-generated dashboards and real-time KPIs.

PLATFORM & INFRASTRUCTURE

- Cut new-service setup from days to hours by establishing the Terraform + Lambda delivery standard — a uniform repo scaffold, GitLab CI that builds artifacts and deploys on apply, and promotion across AWS accounts — now the default across 80+ backend services.
- Architected the event-driven ingestion platform processing 1M+ events a day — topic-routed pub/sub over AWS IoT Core and Kafka/MSK, fanning out to webhooks, live WebSocket clients, a time-series store, and 100+ DynamoDB tables — cutting operational visibility from minutes to seconds.
- Designed a secure device-management and OTA platform for a fleet of connected machines — automated X.509 certificate provisioning, per-device least-privilege policies, remote firmware delivery, and self-healing configuration reconciliation.
- Own authentication and authorization across HTTP, REST, and WebSocket APIs — Lambda authorizers validating Cognito and Azure AD (Entra) JWTs plus scoped API keys, and a dynamic RBAC engine rolled out log-then-enforce to zero mismatches.
- Kept platform spend flat as event volume grew 4x by choosing serverless over always-on compute — no idle EC2 or container capacity, DynamoDB on-demand by default — tying cost to actual load rather than provisioned capacity.

AI & DATA

- Turned week-long, cross-team investigations into self-serve answers with a production LLM agent platform on Amazon Bedrock — a planner-routed multi-model agent (Claude and Nova models) with a 10-tool registry, a sandboxed code interpreter, human-in-the-loop gating on writes, and NL-to-SQL over Athena.
- Delivered the serverless analytics behind a 3,500+ device fleet dashboard — utilization and idle detection over a state engine, a unified per-device 360 view, and Zendesk / Pega service-desk integrations — then moved its endpoints off full-table scans onto precomputed rollups with async backfill.
- Turned alerts into one-click remediation with closed-loop ChatOps in Microsoft Teams: Adaptive Cards that fire on threshold breaches and failure-rate spikes, with operator buttons calling back into Lambda to remediate the device directly from chat.

TECHNICAL SKILLS

Languages Python, JavaScript / Node.js, SQL

Frontend React 19, MUI, Vite, Next.js, Tailwind, Plotly, Highcharts, WebSocket / MQTT clients.

AWS Lambda, IoT (Core, Greengrass), API Gateway, DynamoDB, Kinesis, RDS, S3, SQS / SNS / SES, EventBridge, Athena, MSK, Cognito, EC2

AI & ML Amazon Bedrock (Claude, Nova), agent orchestration & tool use, NL-to-SQL, prompt engineering, human-in-the-loop design

Data & Observability DynamoDB data modeling, PostgreSQL, Athena, InfluxDB, Grafana, CloudWatch, MQTT pub/sub, Kafka consumers

Infrastructure & Security Terraform, GitLab CI/CD, Jenkins / CloudBees, IAM, Entra ID (MSAL), OAuth 2.0 / JWT, RBAC, X.509 / mTLS

EDUCATION

Rutgers University — B.S. in Computer Science, New Brunswick, NJ May 2021